Applying each level of this framework to an example of my swimming pool filter, where the goal is to optimize power consumption, we can see that:

- Data is raw, isolated facts without context. In my home digital twin, this is a simple, un-interpreted number: 2.8 kW.
- Information adds context to the data. The 2.8 kW reading becomes "the pool pump is drawing 2.8 kW of power at 2:15 PM on a Tuesday". This answers the "what, where, and when" questions, giving the data meaning.
- Knowledge reveals patterns and relationships by analyzing information. The digital twin compares the current reading to its historical data and pre-defined rules. The system knows that this pump typically draws 1.2 kW during normal operation, that power draw above 2.0 kW often indicates the filter is clogged and that the filter was last cleaned 2 months ago. Comparing to historical data, it can also tell that power consumption has been gradually increasing over the past three weeks.
- Wisdom applies knowledge to drive action. The digital twin, using its knowledge, can now recommend a course of action: "Since the pool pump is drawing 133% above normal power, indicating likely filter clogging, inspect the filter and backwash if clogged". In an autonomous digital twin, this could include automatically reducing the pump schedule to reduce power use.

5.1.2 The knowledge engineering challenge

While Ackoff’s pyramid provides a good theoretical framework for understanding how data becomes wisdom, digital twin practitioners face the significant challenge of how do you systematically capture and encode the kind of contextual knowledge that I applied intuitively to diagnose my air conditioner problem? In the earlier example, I effortlessly combined multiple pieces of data—history, manufacturer recommendations, environmental context, and past performance patterns—to reach a maintenance decision. This process felt natural because humans excel at contextual reasoning, but translating this capability into digital systems requires deliberate knowledge engineering.

Tacit knowledge is the undocumented, experience-based understanding that expert operators and technicians carry in their heads. This includes intuitive